

Answer Key — Biology Grade 11

Sub-Strand 1.1: Taxonomy I • Lesson 1: Introduction to Taxonomy and Classification: Why Do Our Groupings Differ?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	C	B	A	C	B	D

USE AFTER: PREDICT PHASE — Predict: individual written prediction and Think-Pair-Share on whether the farmer, fisher and student will make the same groups

Q1. A farmer in Kericho, a fisher at Lake Victoria and a student in Nairobi each sort the same eight organisms: goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito. Which prediction is most reasonable?

- A. Only the student in Nairobi can make correct groups
- B. They may make different groups, because each person may choose a different rule such as habitat, use or size**
- C. They must make the same groups, because the organisms are the same
- D. They cannot make any groups, because the organisms are too different

Correct answer: B. People can sort using different rules (habitat, food use, colour, movement), so different groupings are possible. Predictions are not marked wrong at this stage, but this one fits the idea of different criteria.

USE AFTER: OBSERVE PHASE — Observe: card sorting, Step 1, where the group agrees one rule and sorts the eight cards

Q2. A group sorts the organisms with the rule: 'Can move from place to place on its own.' Which set contains only organisms that clearly fit this rule?

- A. Goat, maize plant, mushroom
- B. Mushroom, algae, bean plant
- C. Goat, chicken, mosquito**
- D. Chicken, bean plant, mosquito

Correct answer: C. A goat, a chicken and a mosquito all clearly move from place to place on their own. Maize, bean plants and mushrooms stay fixed in one place.

USE AFTER: OBSERVE PHASE — Observe: Gallery Walk, finding an organism placed in a different group from your own

Q3. In a card-sorting activity, Group A put the mushroom with the plants and Group C put the mushroom in a different group. Both groups wrote a clear rule. What does this show?

- A. The mushroom cannot be classified at all
- B. The same organism can be placed in different groups when groups use different rules**
- C. Group C cheated, because the mushroom can only go with plants
- D. Both groups must have misread the cards

Correct answer: B. Each group used its own criterion, so the same organism ended up in different groups. This was the key observation from the Gallery Walk.

USE AFTER: OBSERVE PHASE — Observe: Step 4, listing Kikuyu, Luo, Kiswahili and English names for the same organism in a table

Q4. Maize is called 'mahindi' in Kiswahili, and it has different names in Kikuyu, Luo and English. What problem can this cause?

- A. People using different names may not be sure they are talking about the same organism**
- B. Only the Kiswahili name is allowed in science
- C. Names do not matter because everyone already uses the same groups
- D. Maize will grow differently in each community

Correct answer: A. Different local names for one organism make it hard for people in different places to be sure they mean the same living thing.

USE AFTER: EXPLAIN PHASE — Explain: debate on 'Is the group that sorted by usefulness (food, pests, disease) wrong?'

Q5. One group sorted the eight organisms by usefulness: food, pests and disease-causing. Another learner says, 'That group is wrong.' Which response fits what the class found?

- A. The group is wrong, because organisms cannot be both useful and harmful
- B. The learner is right, because only habitat can be used to sort organisms
- C. The group is not wrong, because it followed a clear rule, but its rule is only one of many possible rules**
- D. The learner is right, because usefulness is not a real feature

Correct answer: C. Every group was correct by its own criterion. The puzzle is that many correct groupings exist, so no single one works for everyone.

USE AFTER: EXPLAIN PHASE — Explain: Jigsaw/peer teaching question 'What is classification?'

Q6. Which statement best describes what 'classification' means, as used in this lesson?

- A. Finding the most useful organism
- B. Grouping organisms using chosen criteria (rules)**
- C. Collecting organisms from the environment
- D. Giving every organism a new name

Correct answer: B. In this lesson, classification means grouping organisms using chosen criteria. Different criteria give different groupings.

USE AFTER: MODEL BUILDING PHASE — Model: drawing the initial model with the 'problem' arrow and the 'what we need' box

Q7. At the end of Lesson 1, learners draw an initial model with a 'what we need' box. Which idea best fits what a shared system for classifying organisms should provide?

- A. Only people in Nairobi decide the groups
- B. Only organisms that are useful for food are included
- C. Each community keeps its own names and rules, and nobody changes them
- D. The same names and rules that people everywhere can use to identify and study organisms**

Correct answer: D. The puzzle was that different groupings and names stop people communicating, so a shared system needs common names and rules usable by everyone. The lesson does not yet say what that system is.